

TOPICS

- Views selection
- Alignment of views
- Orthographic writing steps
- Basic dimensioning
- Tangency and intersections



View Selection



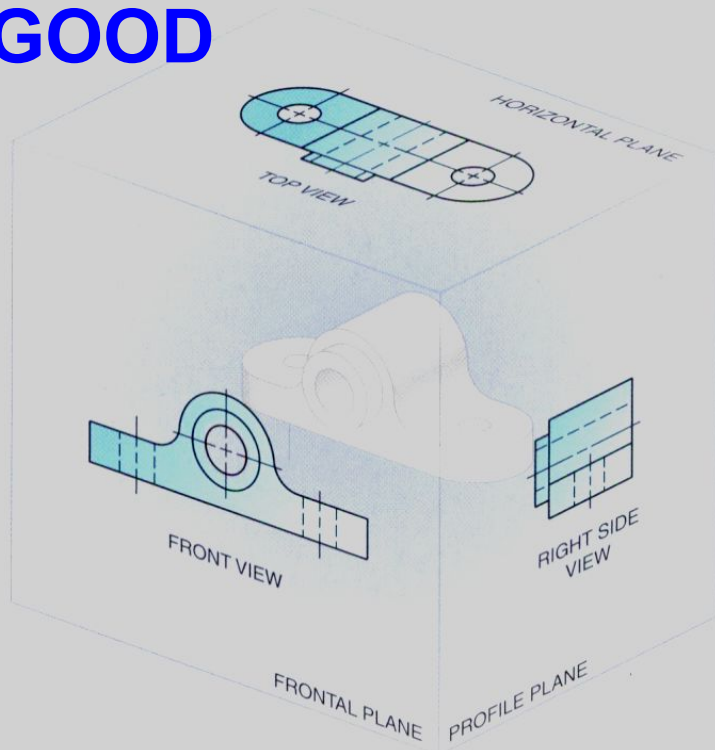
VIEW SELECTION STEPS

1. Orient the object to the best position **relative** to a glass box.
2. Select the front view.
3. Select adjacent views.

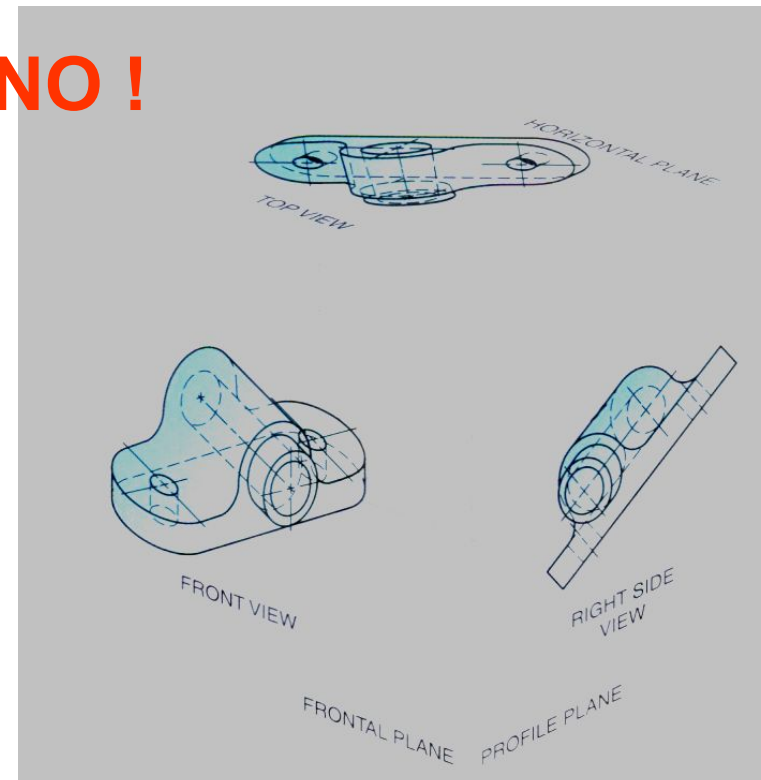
STEP 1 : Orient the Object

- The object should be placed in its **natural position**.
- The object should presents its features in **actual size** and **shape** in orthographic views.

GOOD



NO !



STEP 2 : Select a Front View

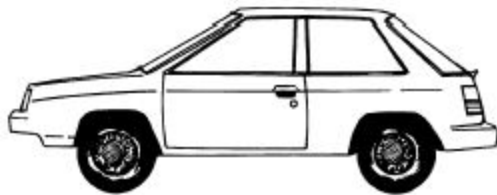
- The object's **longest dimension** should be presented as a **width**.

First choice

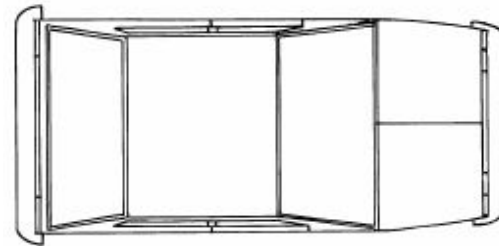


Waste more space

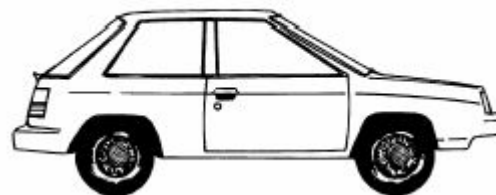
Inappropriate



Second choice



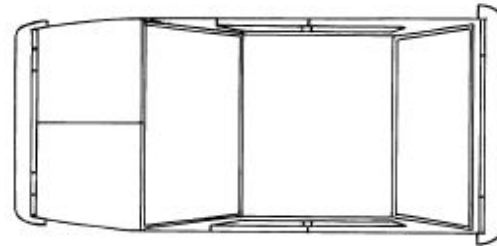
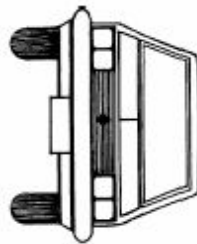
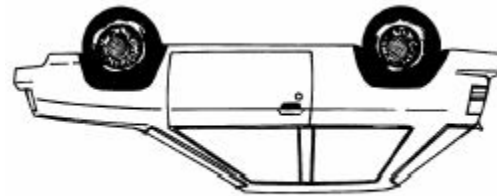
GOOD



STEP 2 : Select a Front View

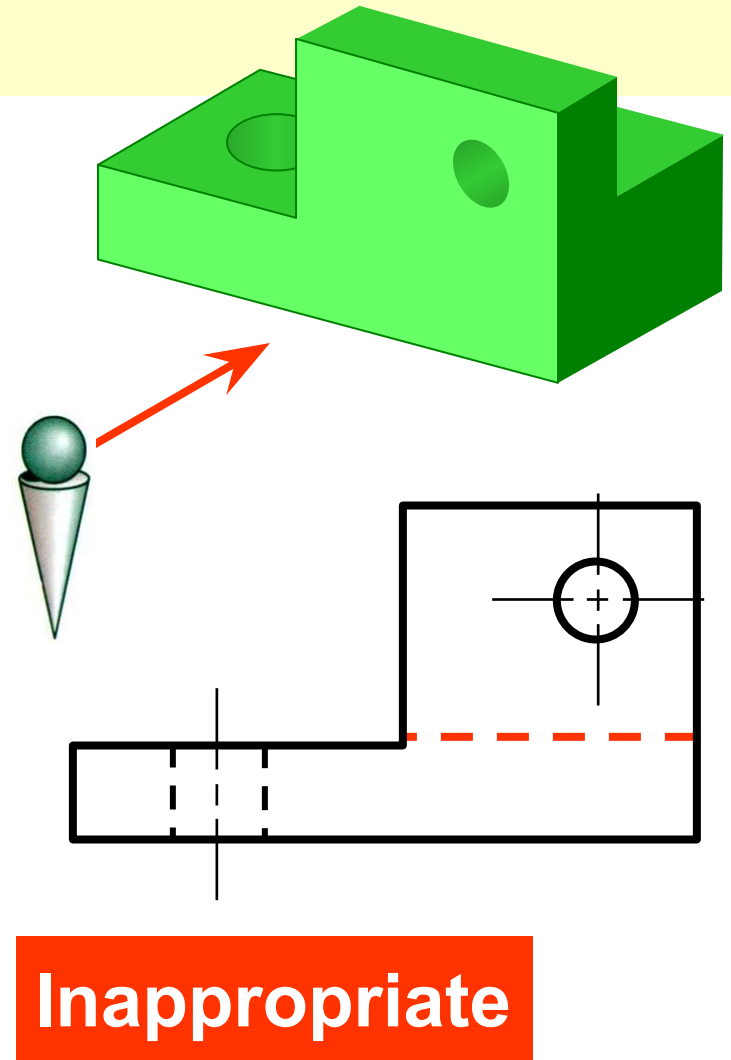
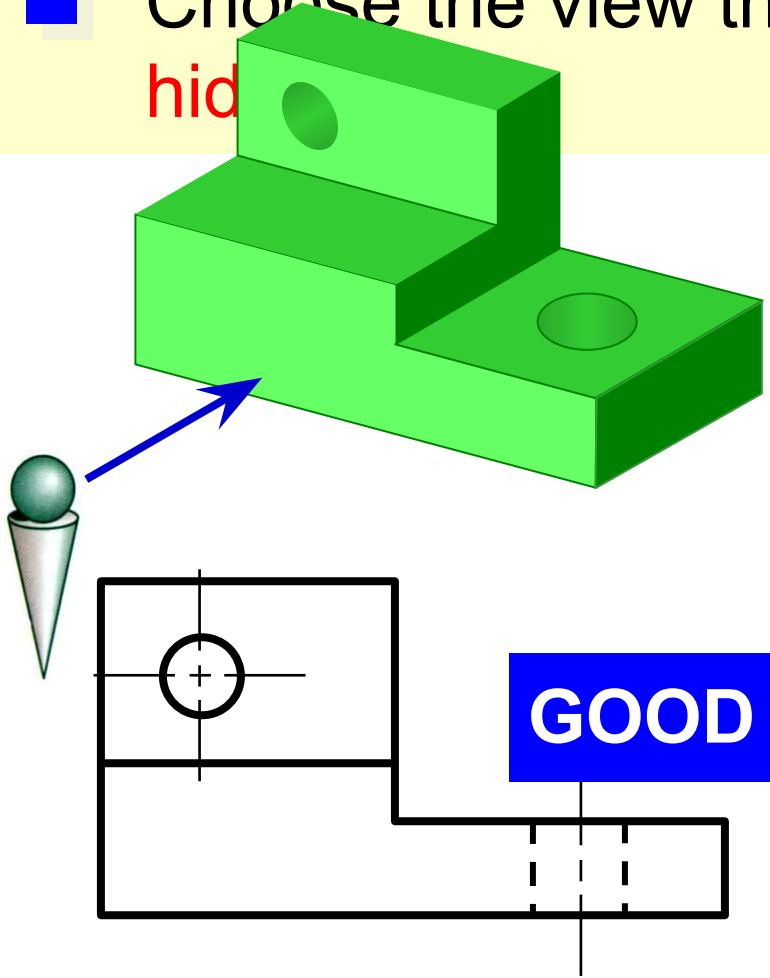
- The adjacent views that are projected from the selected front view should appear in its natural position.

Inappropriate



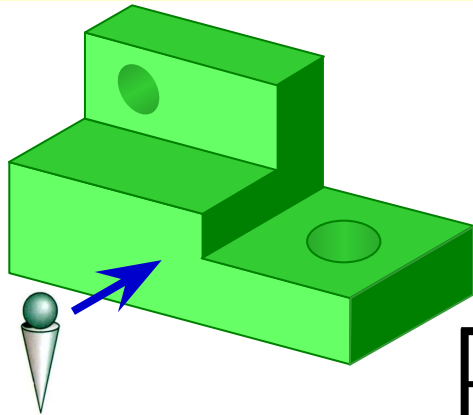
STEP 2 : Select a Front View

- Choose the view that have the **fewest number of hidden lines**

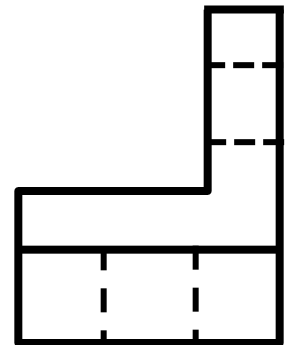
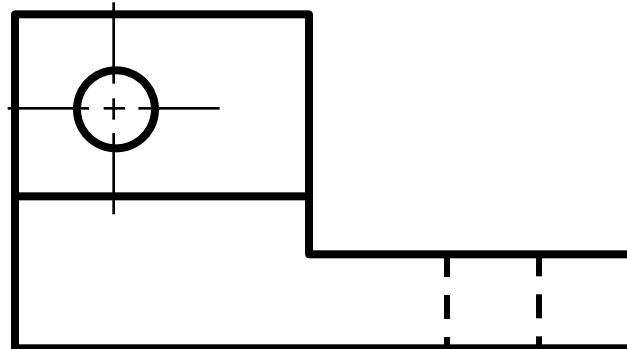
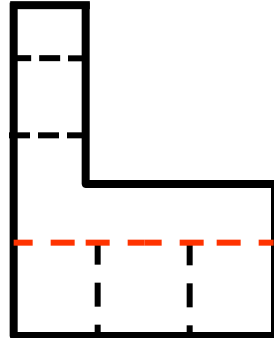
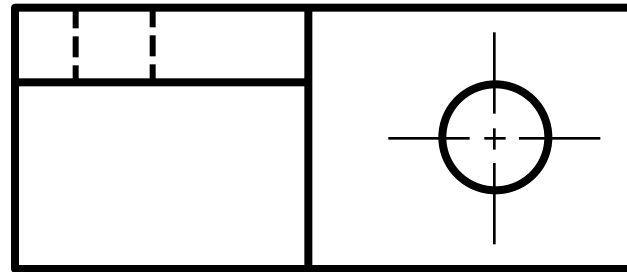


STEP 3 : Select an Adjacent View

- Choose the view that have the fewest number of hidden lines.

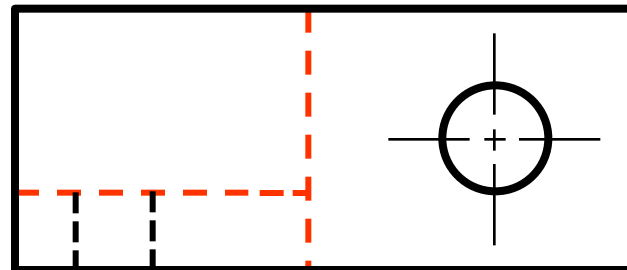


GOOD



Inappropriate

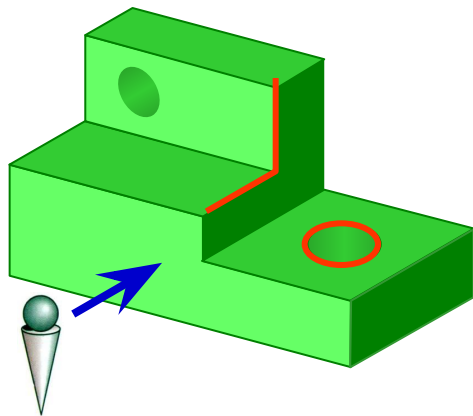
Inappropriate



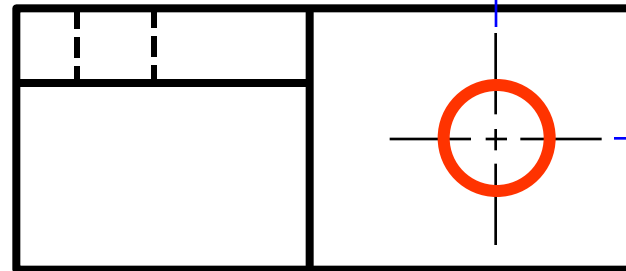
GOOD

STEP 3 : Select an Adjacent View

- Choose the **minimum** number of views that can represent the major features of the object.

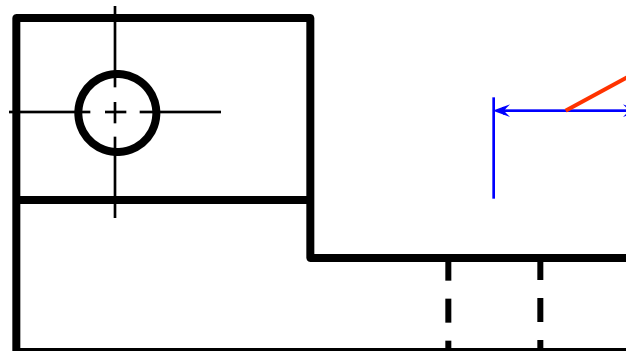


Necessary

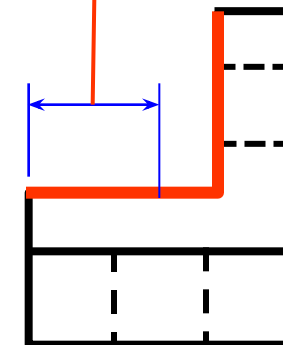


Hole's location can be specified on the same view.

Easy to understand



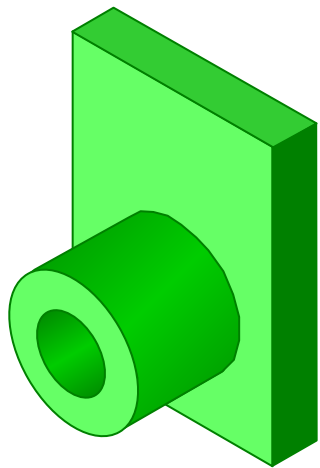
Difficult to interpret.



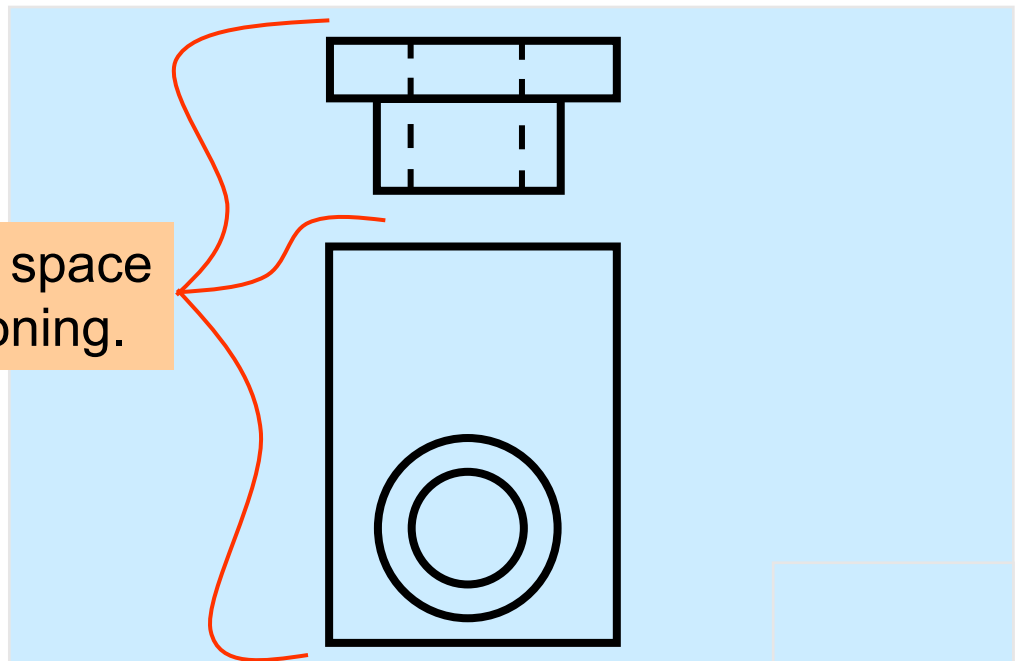
Necessary

STEP 3 : Select an Adjacent View

- Choose the views that are suitable to a drawing space.



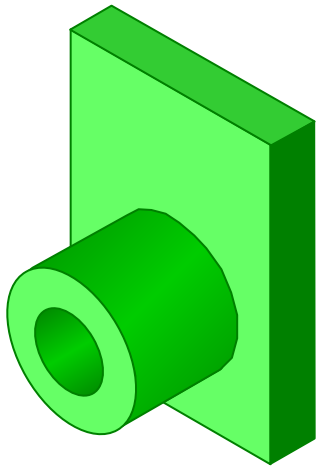
Not enough space
for dimensioning.



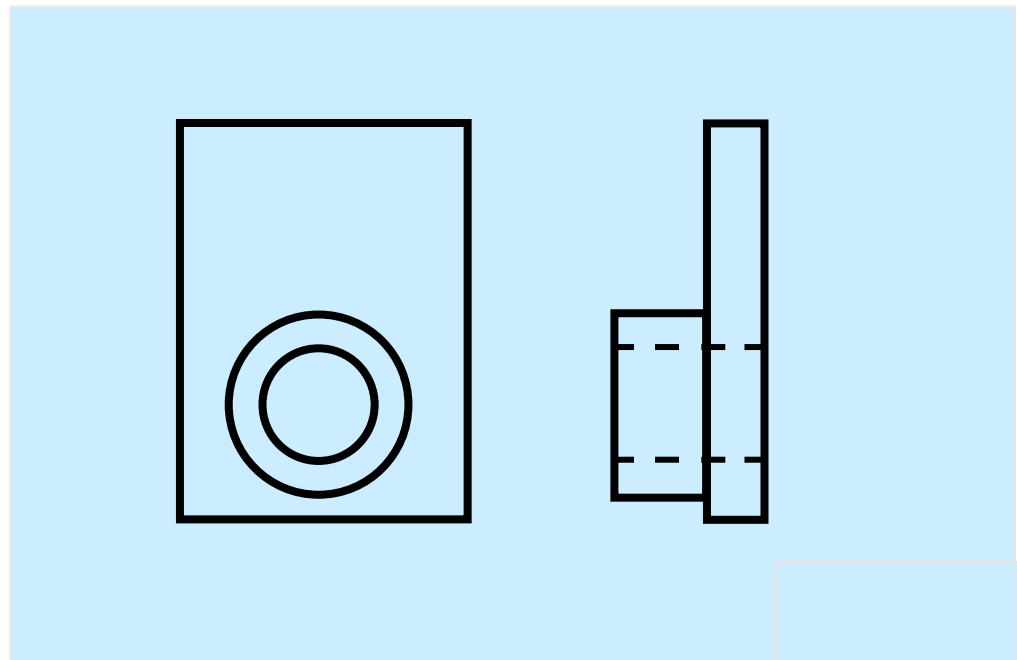
POOR

STEP 3 : Select an Adjacent View

- Choose the views that are suitable to a drawing space.

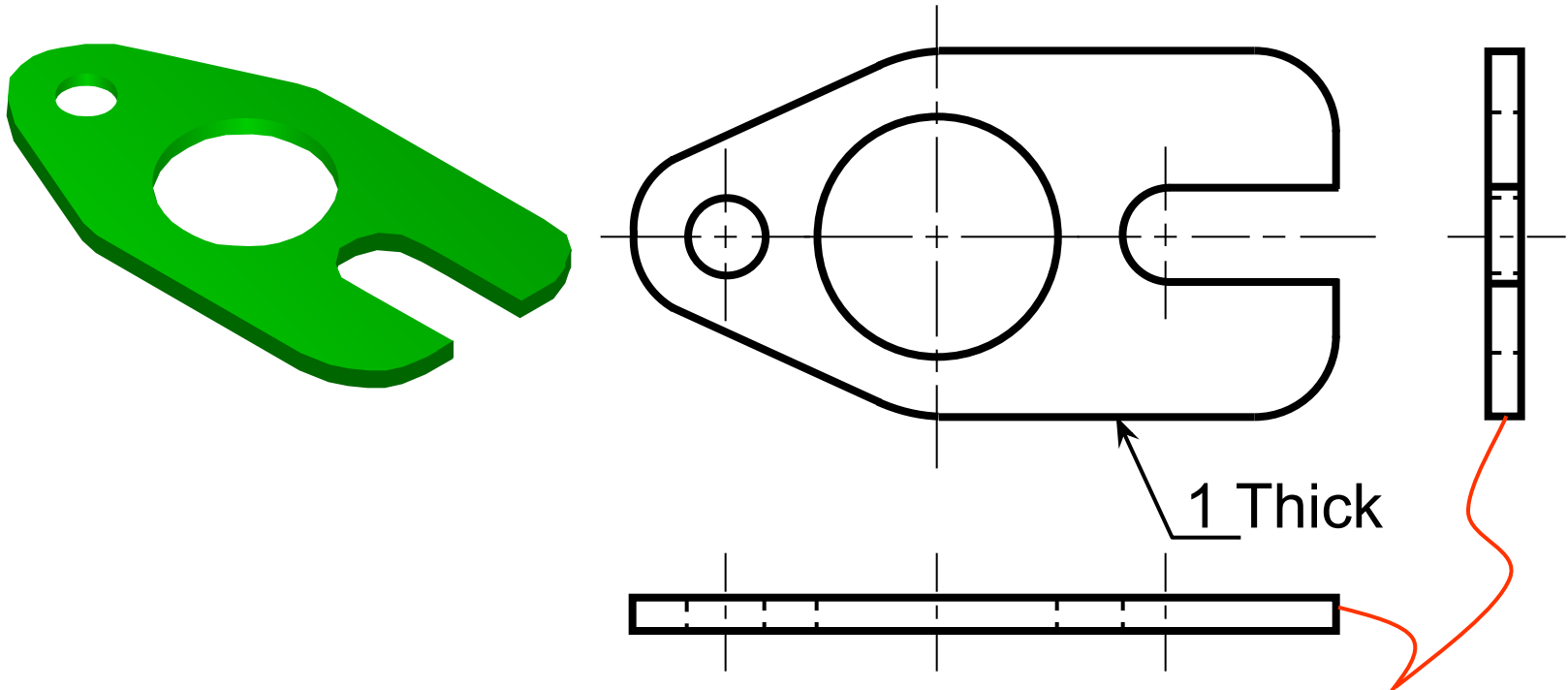


GOOD



ONE-VIEW DRAWING

- ***Flat part*** having a uniform thickness.

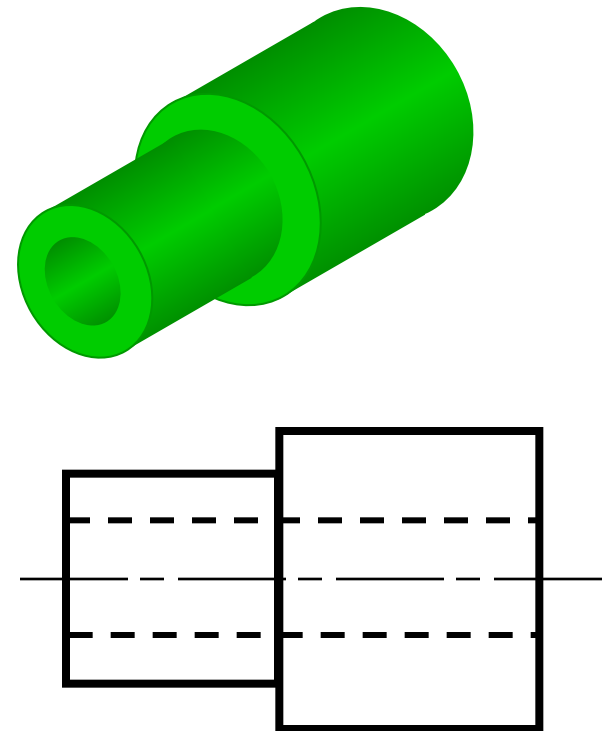
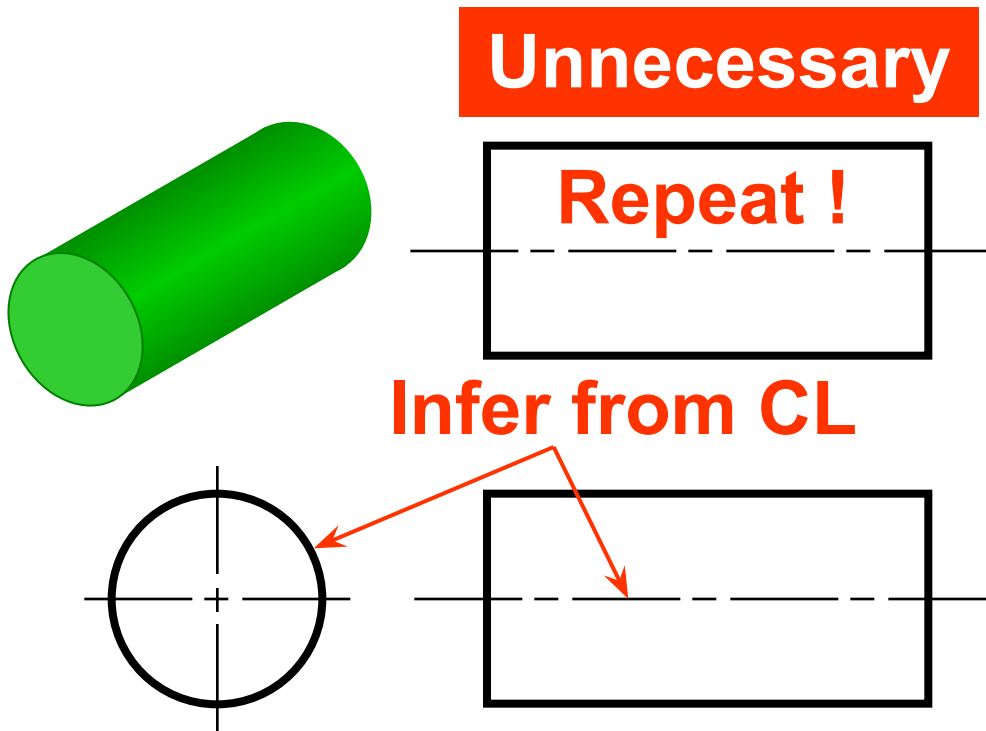


Unnecessary

These 2 views provide only information about the part thickness !

ONE-VIEW DRAWING

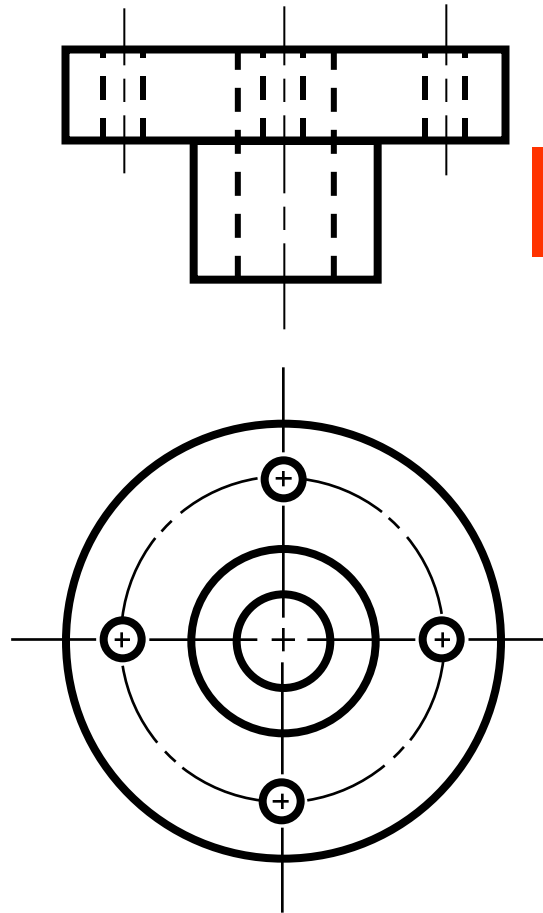
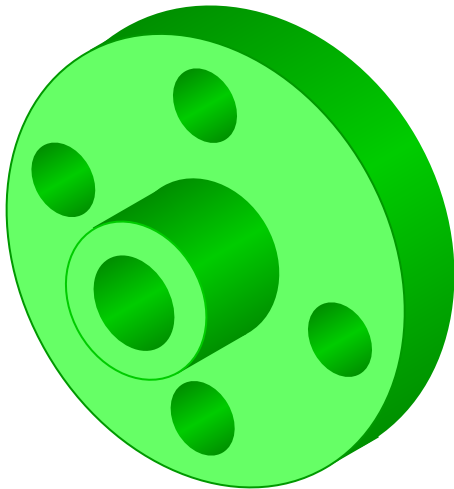
■ *Cylindrical-shaped* part.



Unnecessary

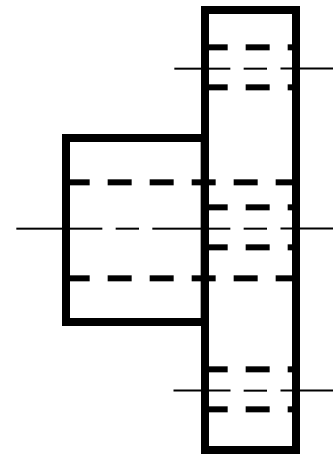
TWO-VIEW DRAWING

- There exists an identical view.



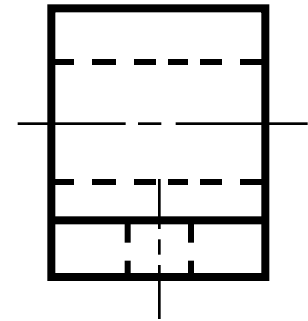
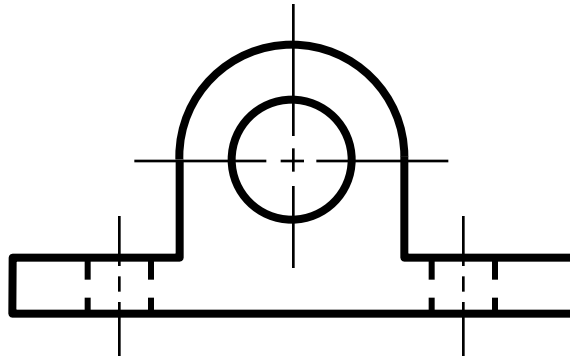
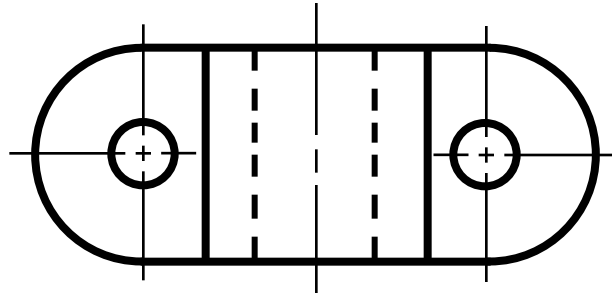
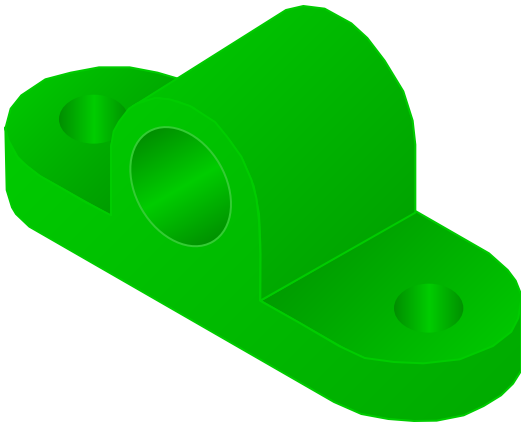
Repeat !

Unnecessary



TWO-VIEW DRAWING

- The 3rd view has no significant contours of the object.



Unnecessary



Alignment of Views



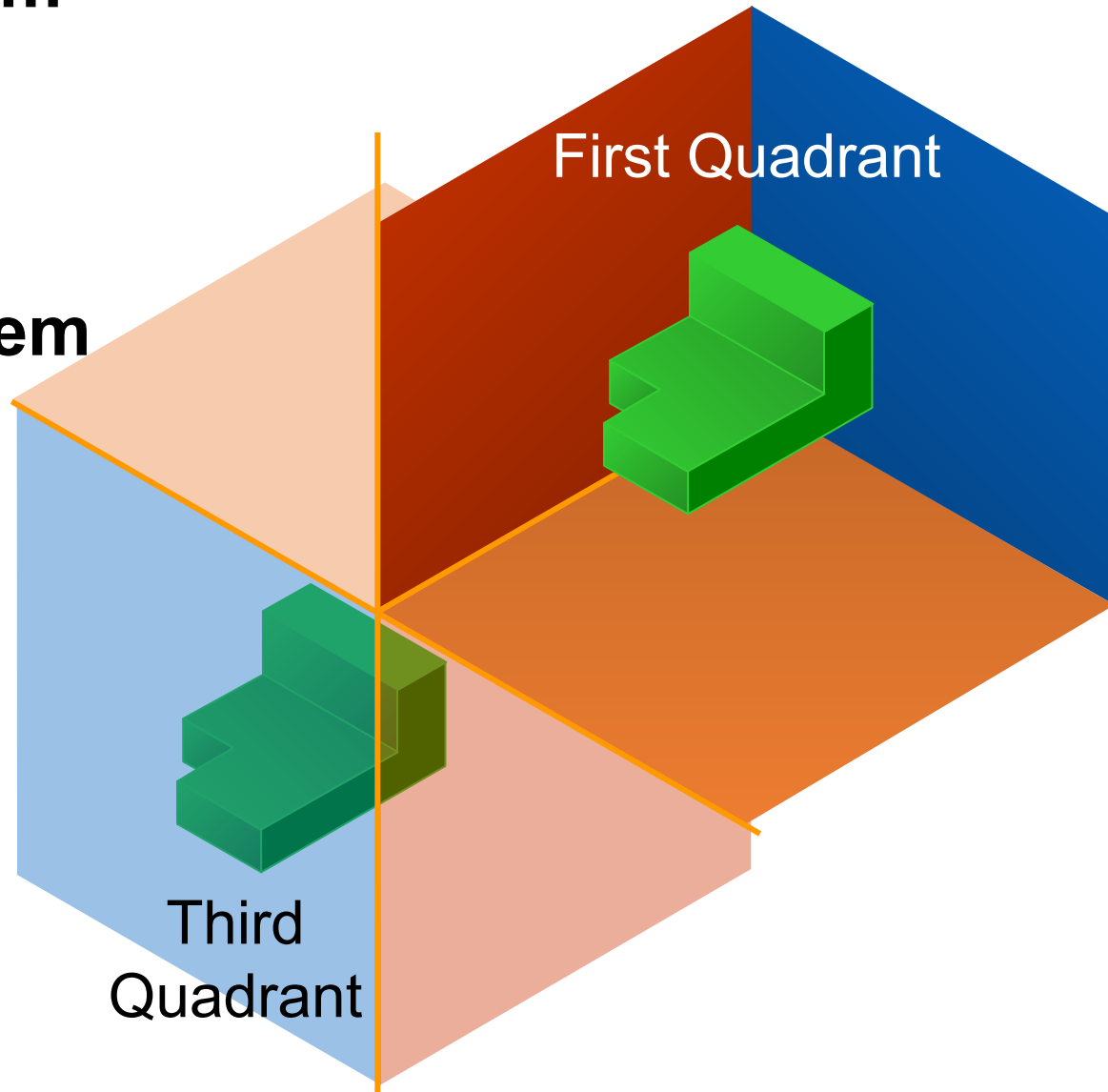
PROJECTION SYSTEMS

1. **First** angle system

- European country
- ISO standard

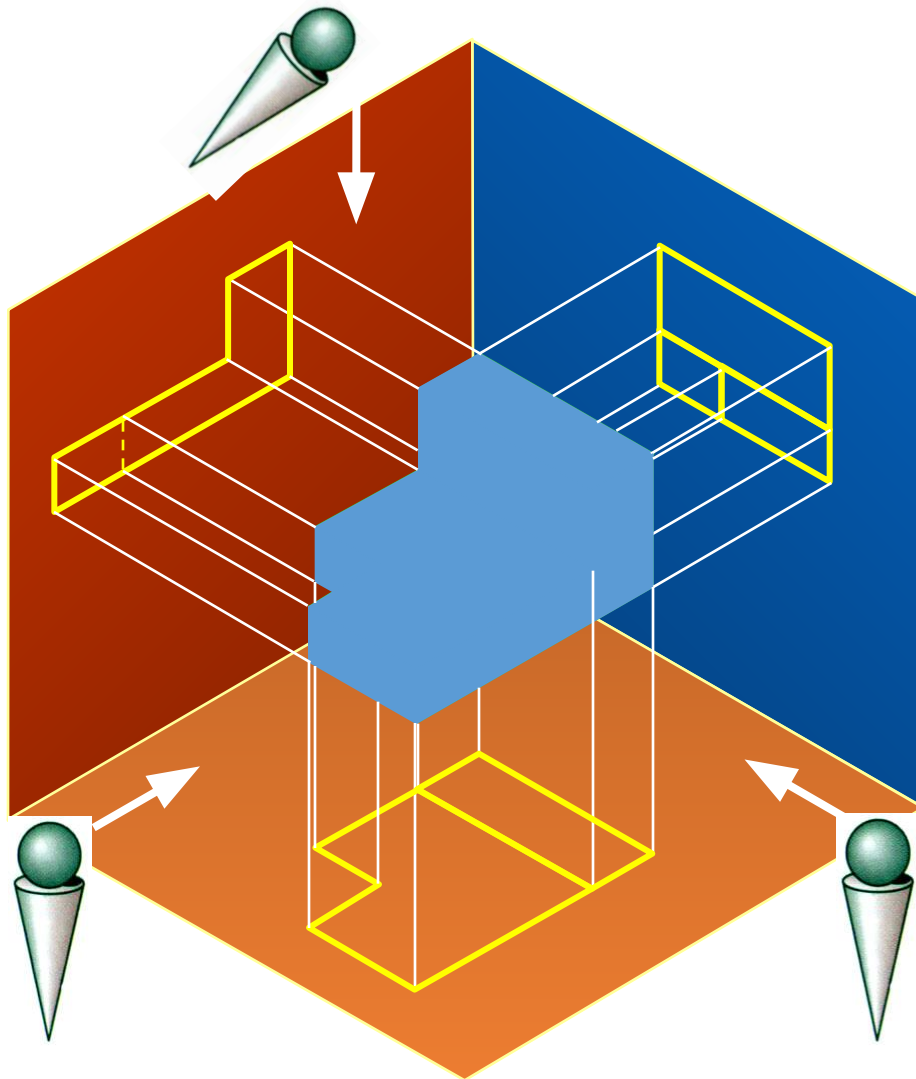
2. **Third** angle system

- Canada, USA, Japan, Thailand

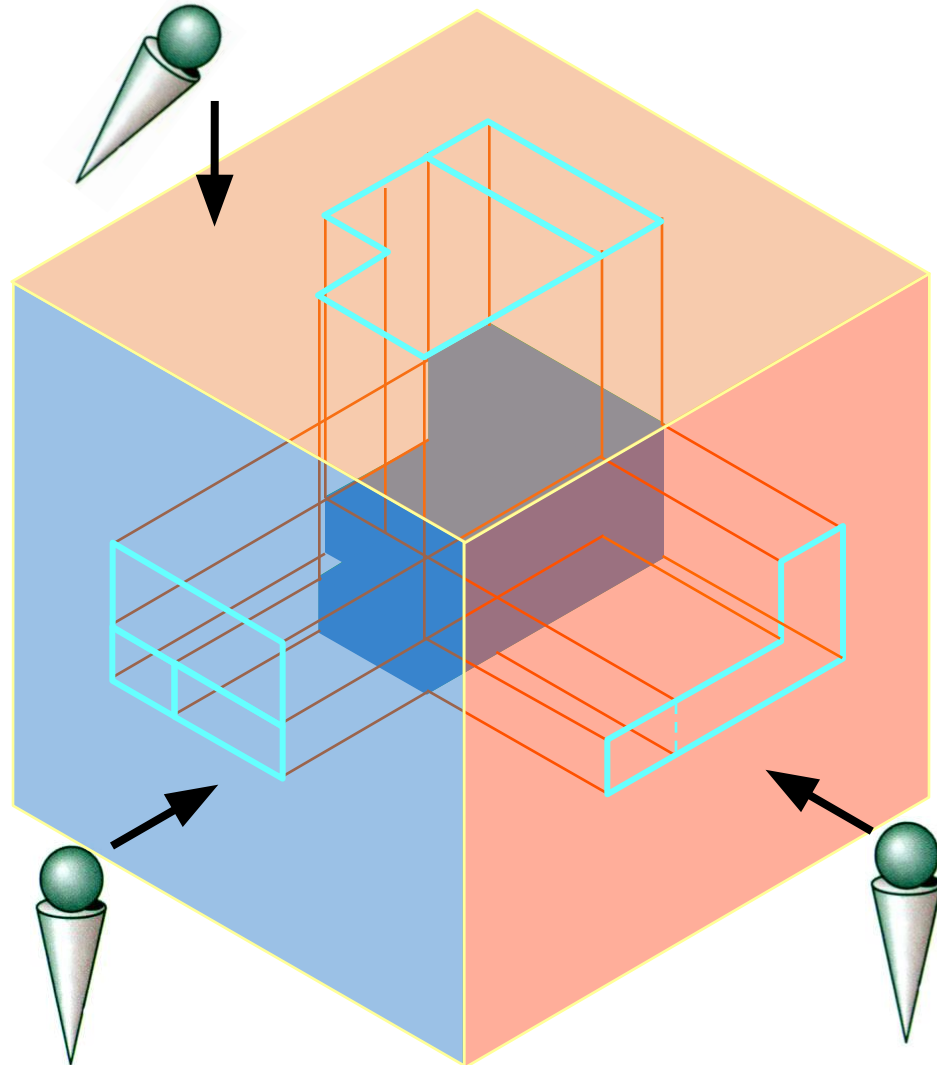


ORTHOGRAPHIC PROJECTION

1st angle system

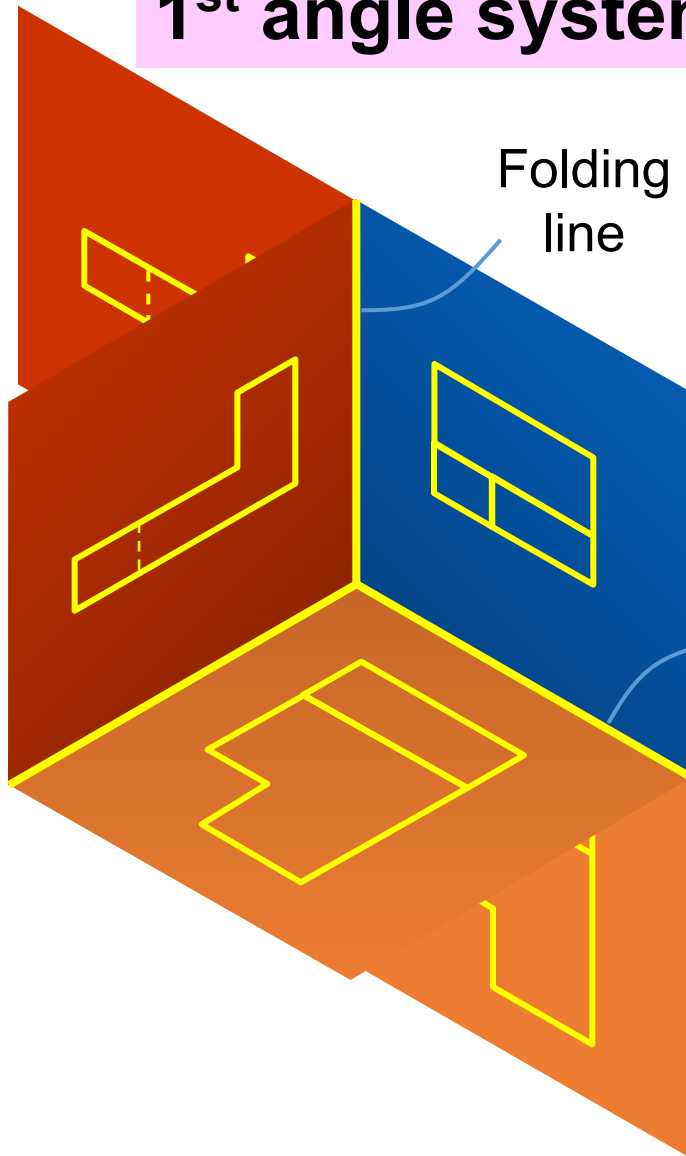


3rd angle system

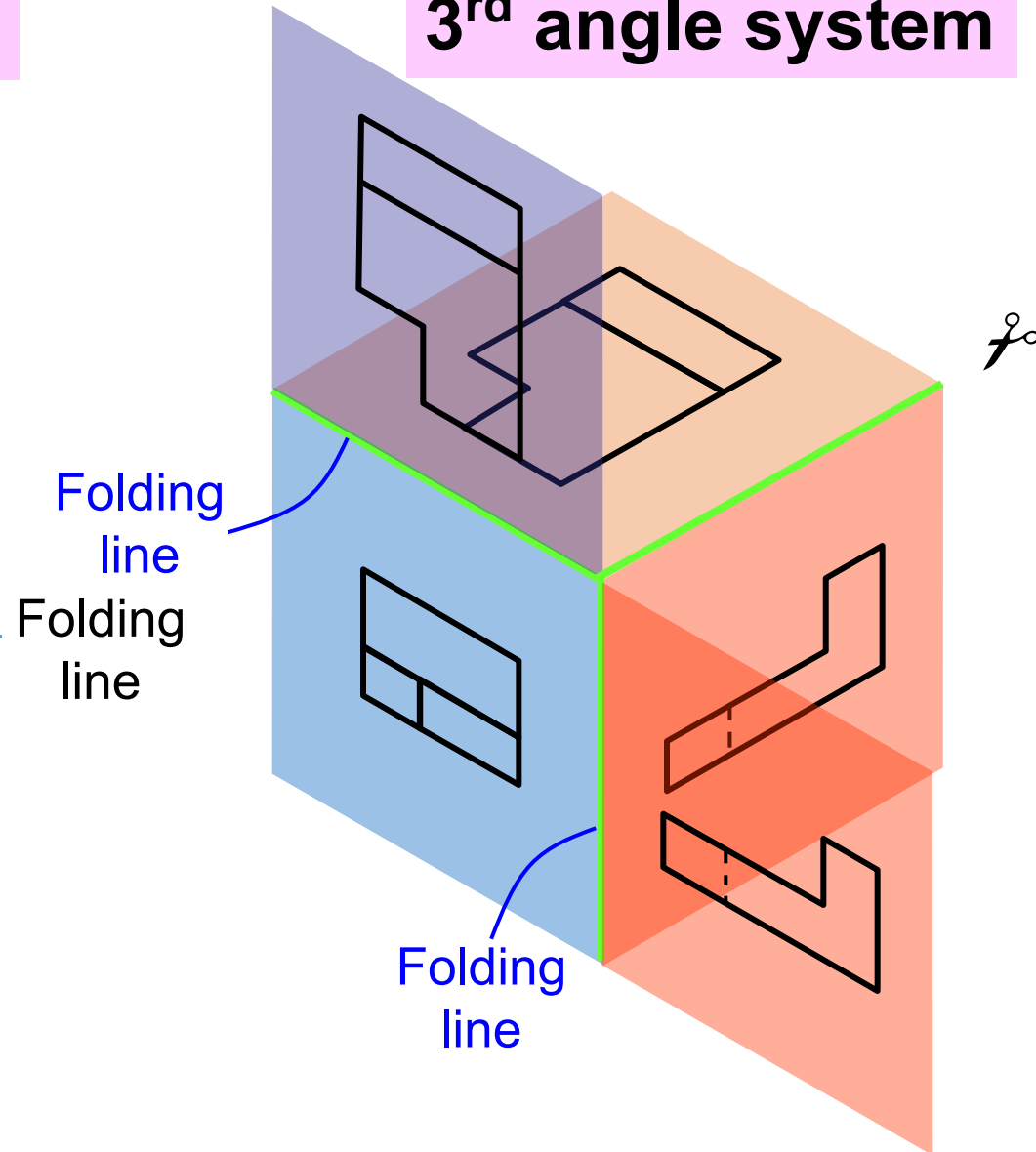


ORTHOGRAPHIC VIEWS

1st angle system

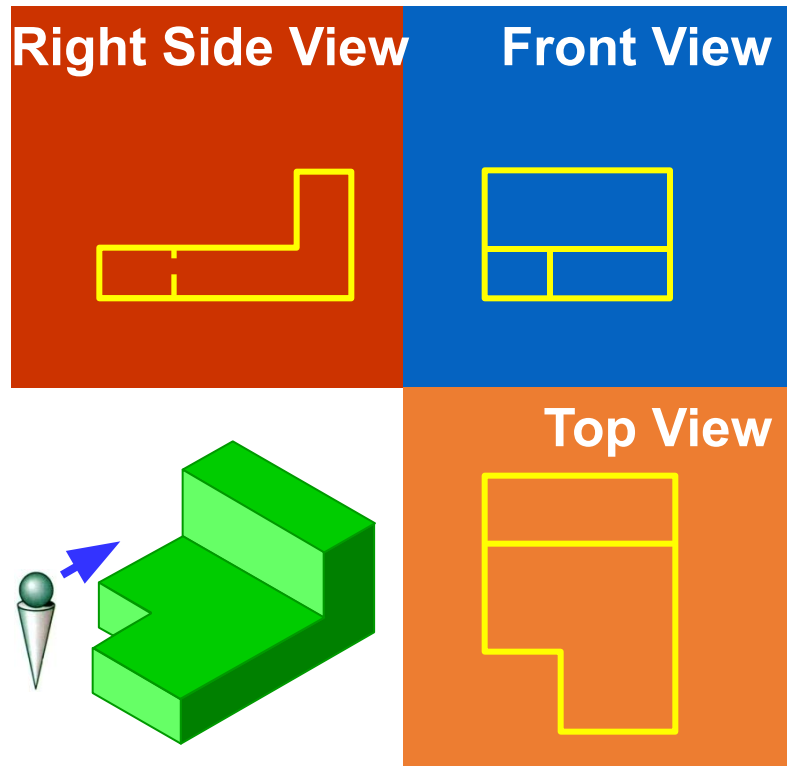


3rd angle system

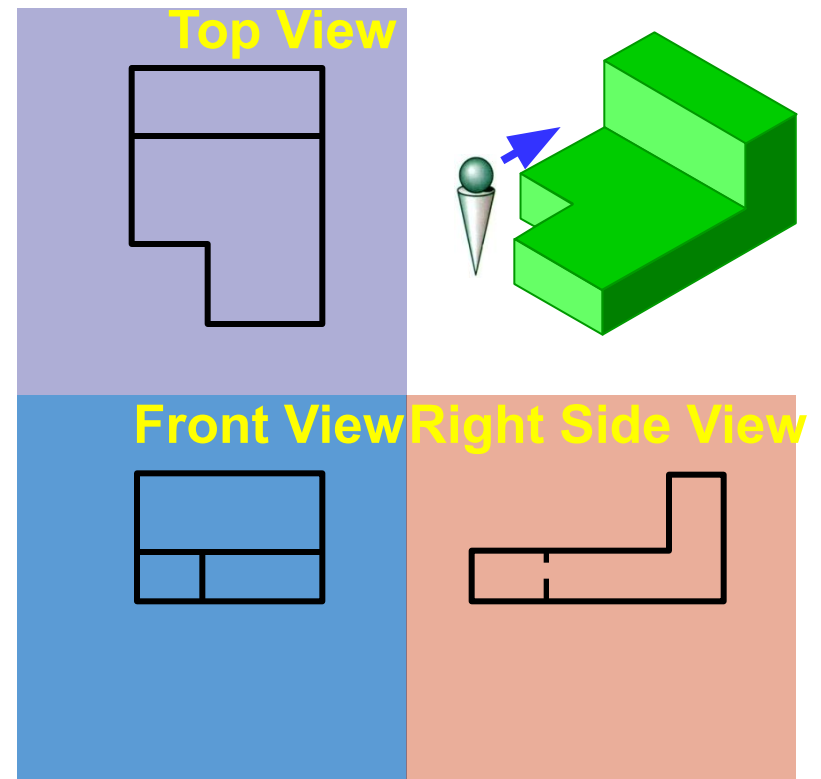


ORTHOGRAPHIC VIEWS

1st angle system

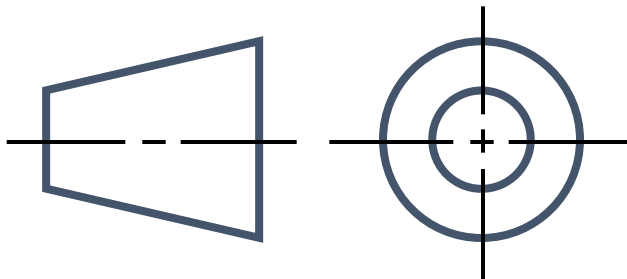
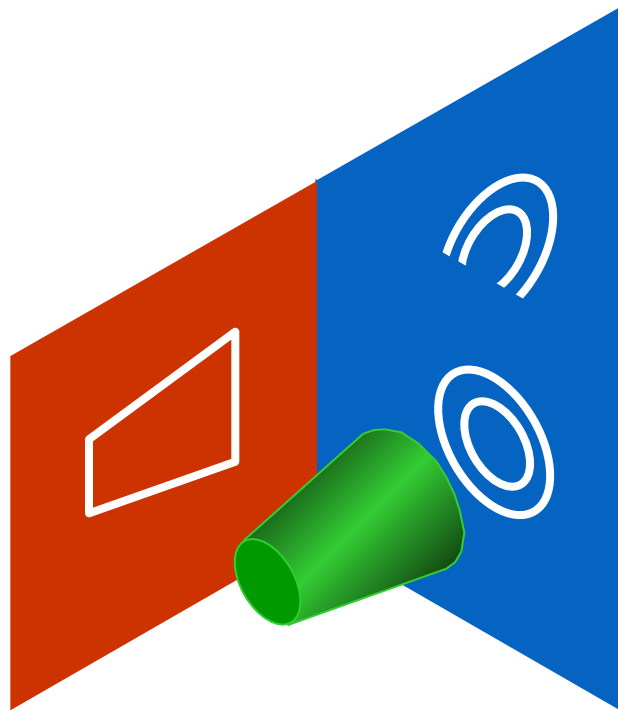


3rd angle system

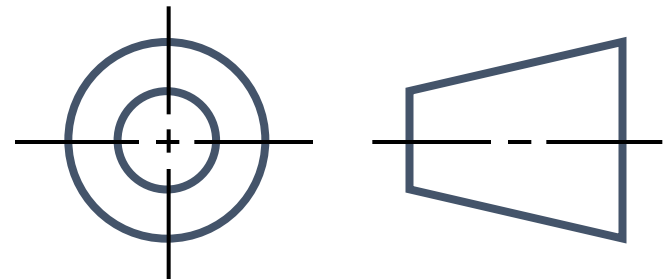
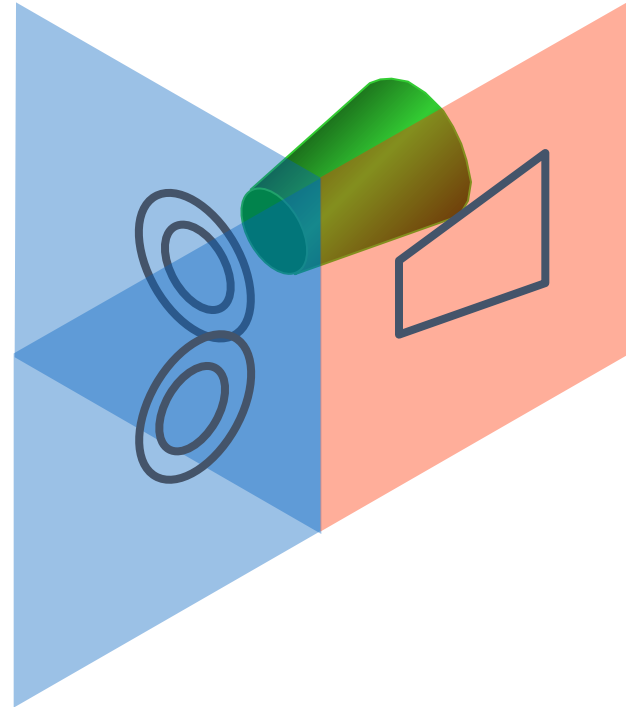


PROJECTION SYMBOLS

First angle system

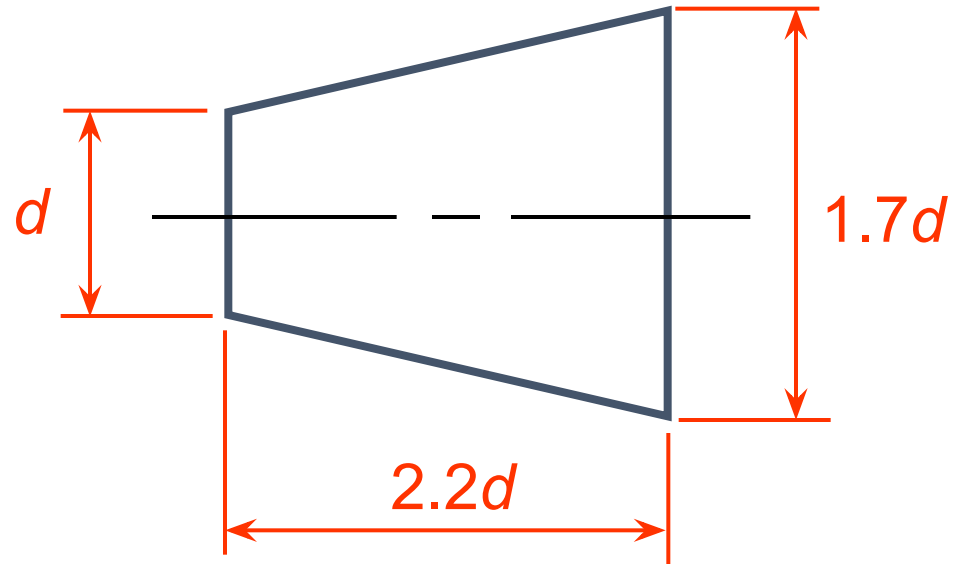
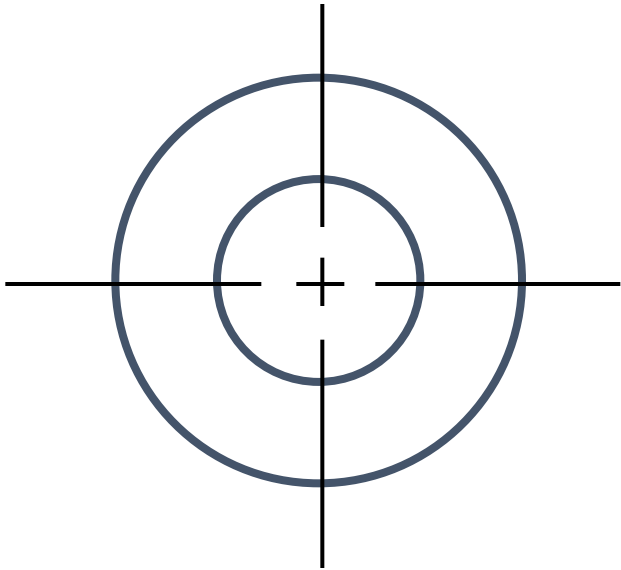


Third angle system



PROJECTION SYMBOLS

Suggested proportion





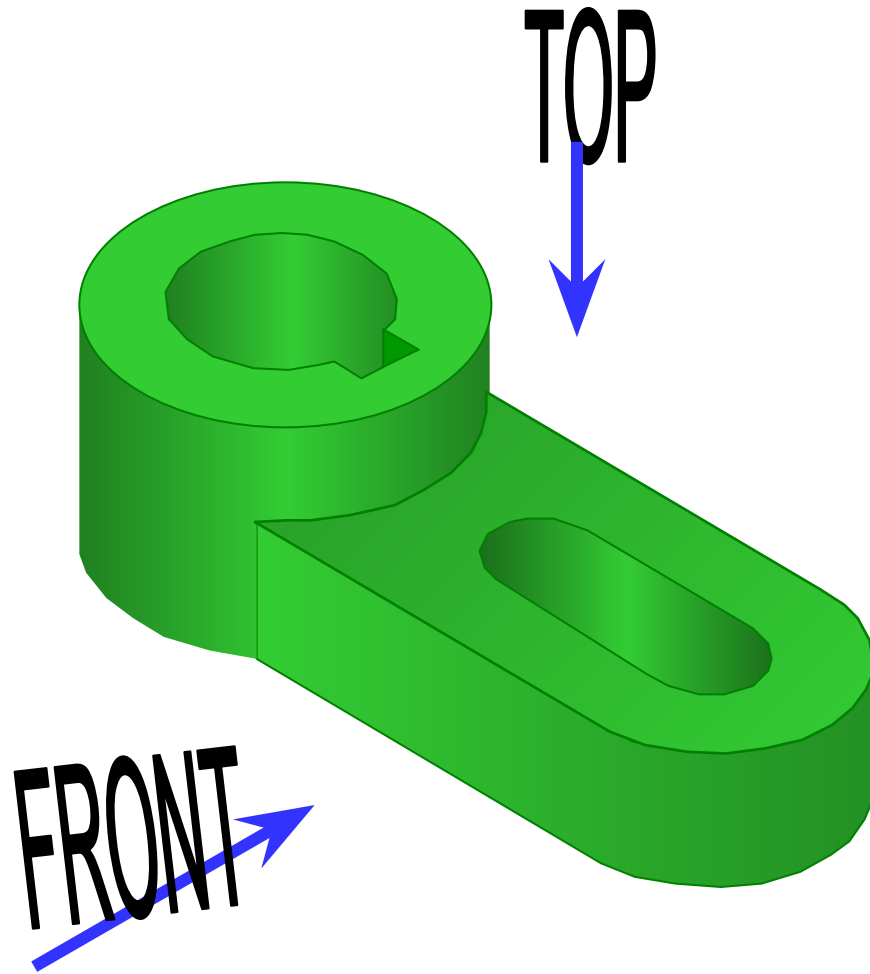
Orthographic Writing Steps



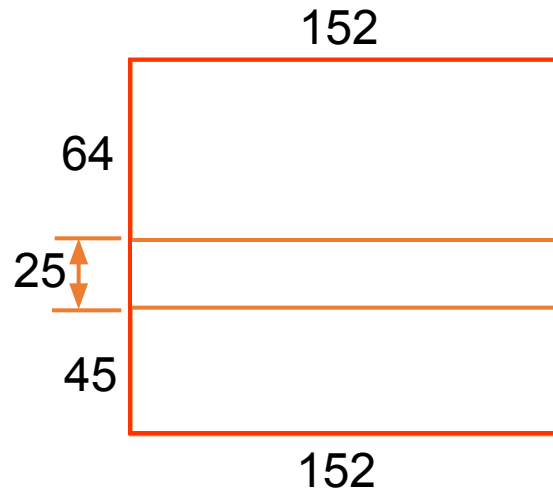
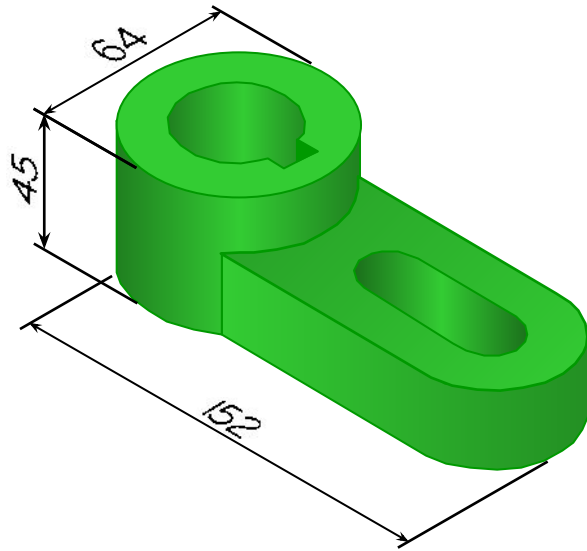
WRITING STEPS

1. Select the necessary views
2. Layout the views.
3. Project the views.
4. Dimension the views.

1. SELECT THE NECESSARY VIEWS



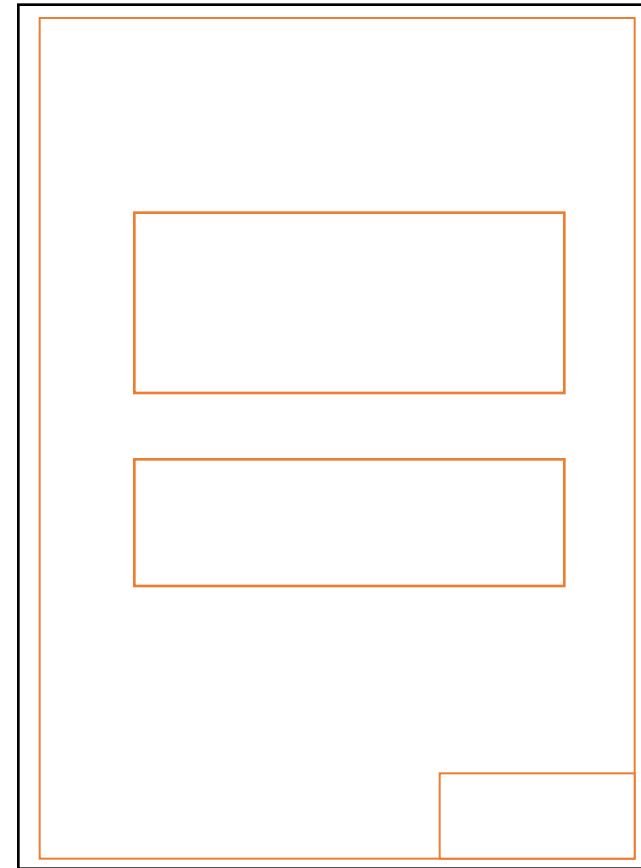
2. LAYOUT THE VIEWS



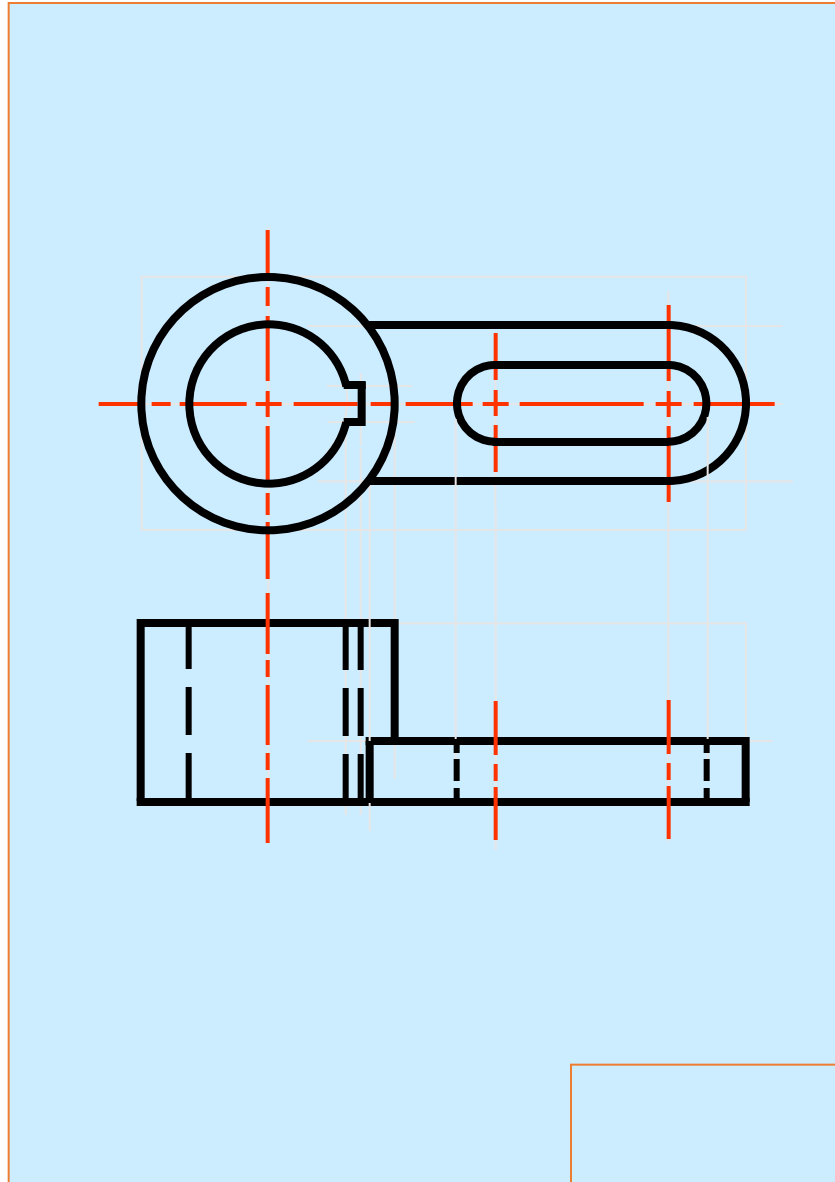
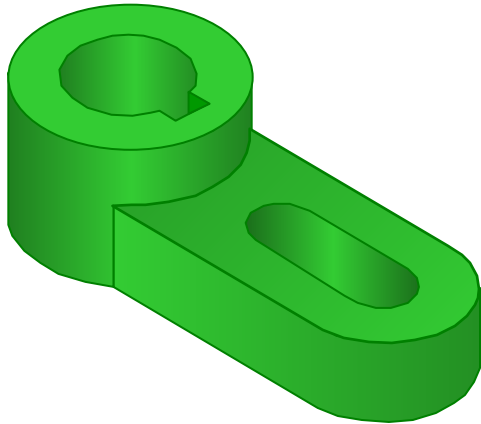
Choose an appropriate scale

1:1

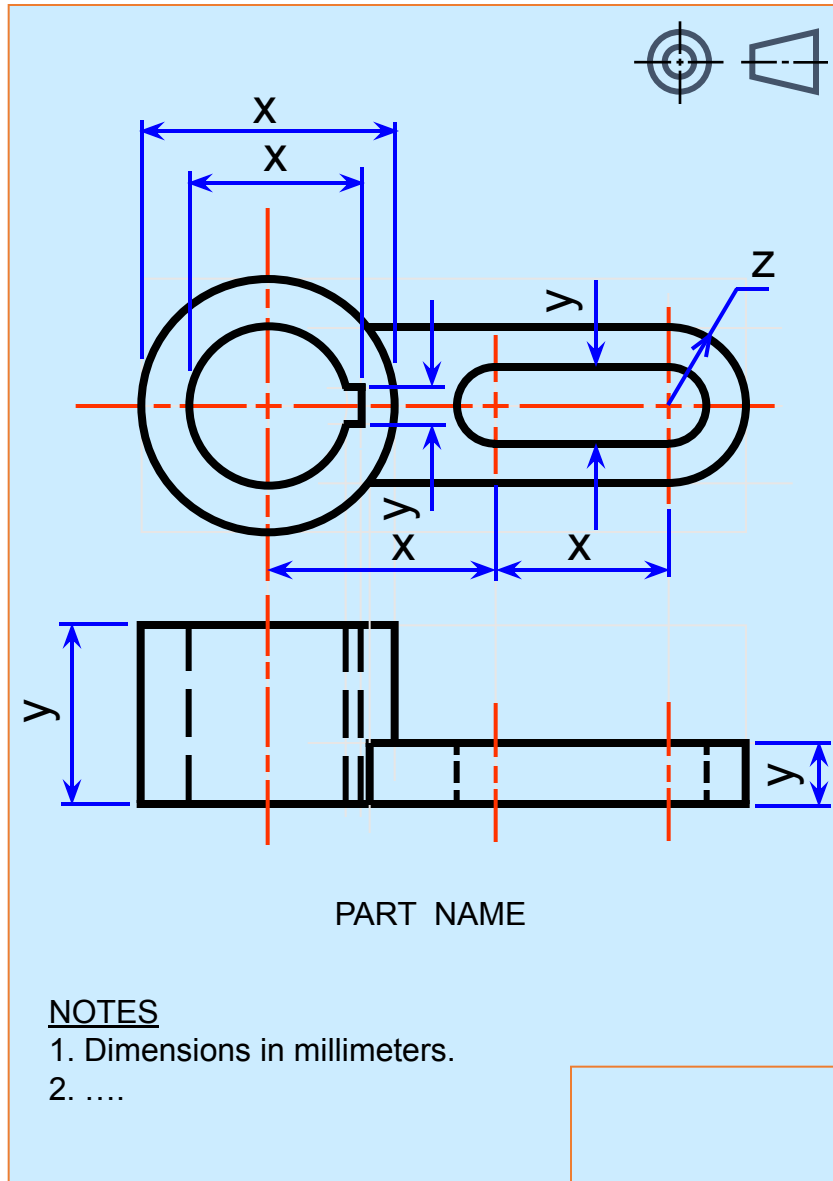
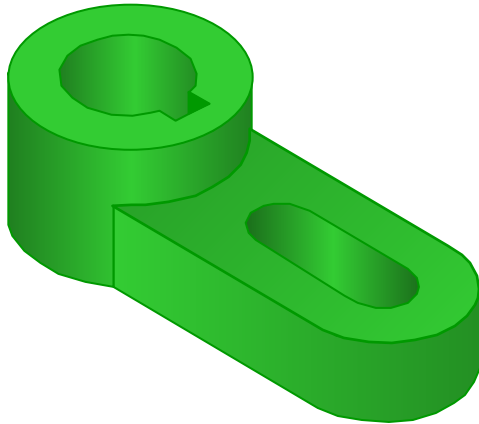
A4



PROJECT THE VIEWS

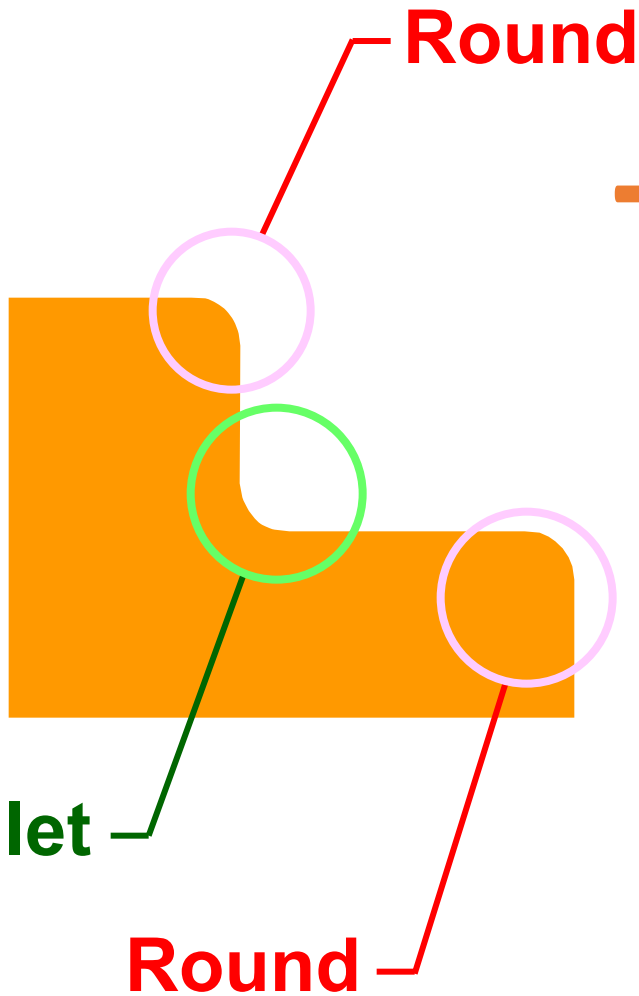


DIMENSION THE VIEWS



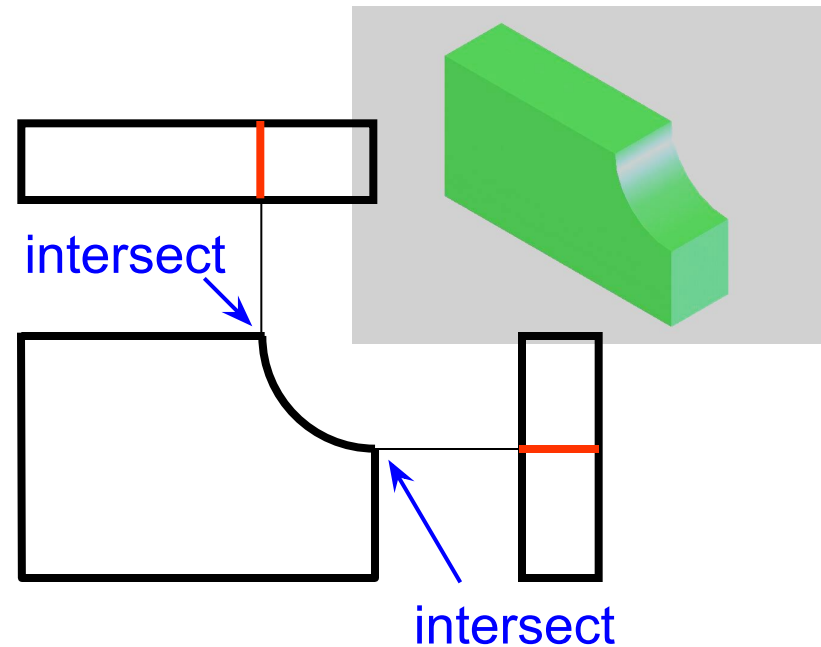
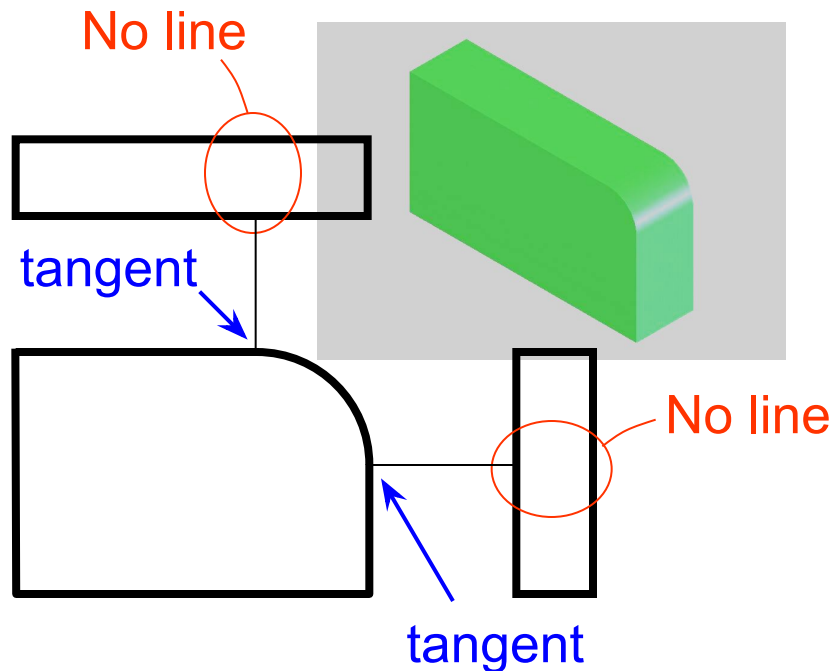


Tangencies and Intersections

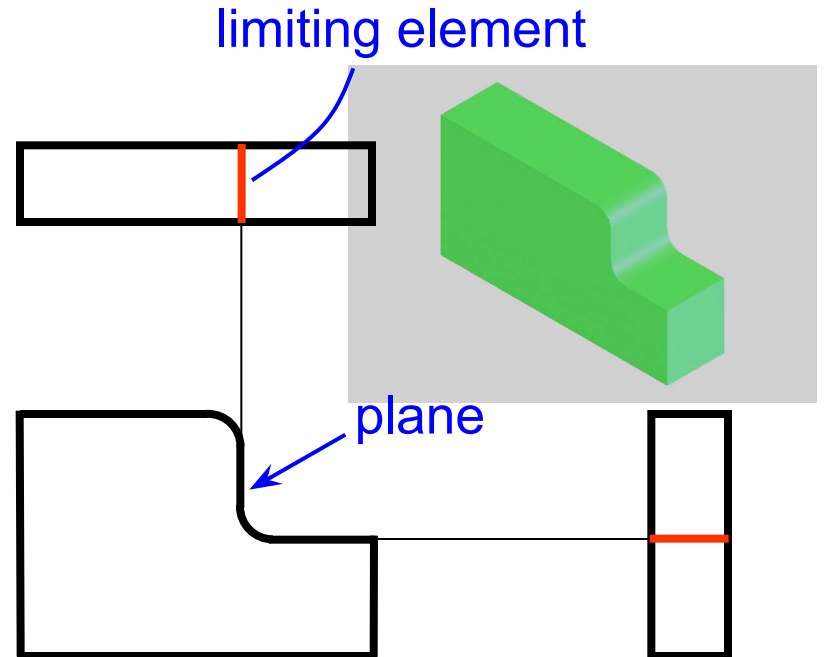
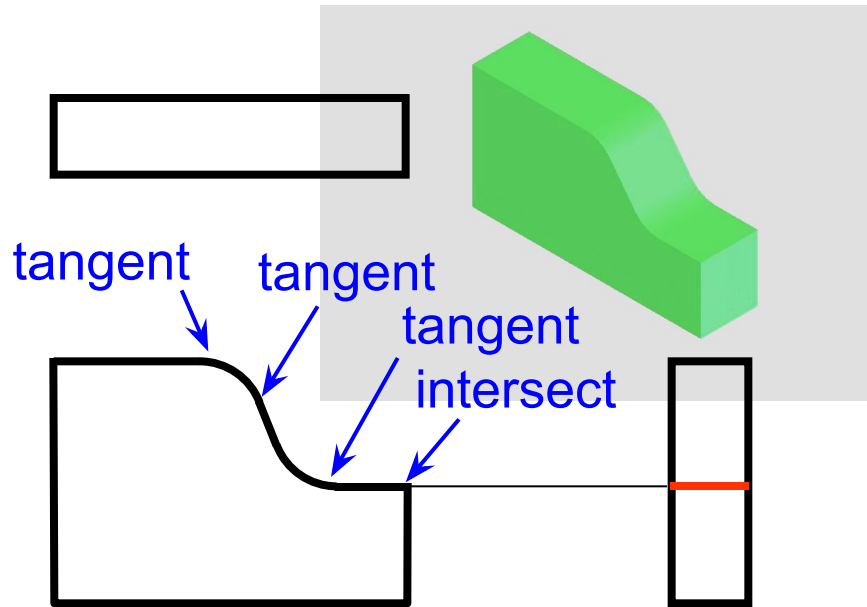


TANGENT & INTERSECTION

- No line is formed when curved surface *tangent* to a plane surface.
- Line is formed when curved surface *intersects* a plane surface.



TANGENT & INTERSECTION



TANGENT & INTERSECTION

